

The Umwelt-Frontier Orchestrator: Pattern, Mechanism, and Preregistered Adversarial Validation

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Abstract

This paper specifies the Umwelt-Frontier Orchestrator (UFO), a multi-agent orchestration pattern for wide parallel synthesis tasks. UFO combines three commitments: role-specialized peers that generate heterogeneous proposals, strict Pareto dominance as the only admissibility filter, and the absence of a pre-specified terminal utility. The resulting mechanism, which we call *exclusion-topology*, does not rank all proposals against a final target. Instead, it maintains a boundary of inadmissible moves: a candidate is admitted only when it harms no declared axis and improves at least one. The paper separates pattern-level claims from evidence-level claims. The pattern layer defines the rule set, boundary mechanics, lineage, and failure modes. The evidence layer reports preregistered internal validation designs: single-node ablations, pair-removal probes, mutation tests, and a restricted memory-substrate comparison. The current evidence is therefore best interpreted as specification-internal validation rather than external benchmark proof. We close with preregistered hypotheses, limitations, and a milestone plan for comparable task-level evaluation.

Keywords: multi-agent orchestration; strict Pareto dominance; exclusion-topology; axis governance; no-goal search; mutation testing; auditability.

1 Introduction

Multi-agent orchestration is often described as a scheduling problem: assign roles, route messages, collect outputs, and summarize. In many synthesis domains, however, coordination is also a *move-acceptance* problem. Many candidate moves can be generated in parallel; each candidate may improve one value while degrading another; and the relevant axes may be incomplete, delayed, or hard to collapse into a scalar without hiding safety-relevant trade-offs.

The Umwelt-Frontier Orchestrator (UFO) addresses this setting by replacing scalar choice with explicit admissibility. A proposed move is accepted only if it is non-regressive over a declared axis set and strictly improves at least one axis. The orchestrator therefore does not ask whether a move maximizes a total score. It asks whether the move crosses a boundary of admissibility without damaging any protected dimension.

This framing is intentionally narrow. UFO is a pattern for tasks where broad proposal pressure is useful, such as incident-response-style synthesis, design review, adversarial validation, or other workflows in which multiple independent dimensions must remain visible. It is not claimed as a universal replacement for graph-native workflow systems, fixed-horizon debate, or single-agent reasoning. Recent work on multi-hop reasoning is treated here as a scope warning rather than a

contradiction: sequential or contained reasoning can favor single-model approaches when thinking-token budgets are controlled [8].

The contribution is twofold. First, we give a compact formal specification of UFO as a compositional pattern: heterogeneous proposal generation, strict Pareto admissibility, externally declared axes, and saturation-based exit. Second, we introduce *exclusion-topology* as the mechanism name for this design. The term marks a boundary process: the orchestrator defines what is inadmissible, lets proposal pressure explore the feasible interior, and stops only when no pending proposal crosses the boundary.

Claim discipline. The paper separates the *pattern layer*, which defines the mechanism, from the *evidence layer*, which records preregistered probes and internal consistency checks. The current paper does not claim end-to-end superiority against other orchestration systems. External comparisons require comparable substrates, task regimes, budgets, and observables.

2 Background and Related Design Context

UFO assembles ideas already familiar in multi-objective optimization, quality-diversity search, cognitive framing, and human-centered governance. The claim is not that these ideas are new in isolation. The claim is that their integration as an admissibility-first orchestration loop has a distinct failure profile.

2.1 Admissibility instead of scalar collapse

Scalar scoring is convenient but brittle when an environment exposes delayed harms or hidden dependencies. Strict Pareto dominance blocks proposals that trade away any declared axis, which preserves option space while still allowing monotone movement. This connects UFO to classical Pareto dominance [1] and to operational multi-objective methods such as NSGA-II [2]. UFO does not import the full evolutionary algorithm. It imports the dominance gate and relocates it to the orchestration layer.

2.2 No-goal search and illumination

Novelty search showed that objectives can be deceptive: optimizing directly toward a target can trap search in local optima [3]. MAP-Elites reframed search as illumination of a structured behavior space rather than extraction of a single best point [4]. UFO adopts the anti-collapse lesson while keeping explicit axes: it has no terminal scalar utility, but it still rejects moves that regress any declared dimension.

2.3 Axis governance as Umwelt discipline

The term *Umwelt* is used as a methodological shorthand rather than a biological identity claim. Von Uexküll’s notion of organism-specific worlds [9] and de Waal’s critique of human-shaped competence tests [10] motivate a design warning: if the axis set reflects evaluator convenience rather than task structure, the filter may be rigorous over the wrong map. UFO therefore treats axes as external governance inputs declared before orchestration begins.

2.4 Legibility and metacriteria

Formal filters create auditability, but excessive legibility can erase local practical knowledge. Scott’s account of legibility and metis [14] and Pirsig’s distinction between recognized quality and formal definition [13] motivate a second warning: the axis set is provisional. Recognition and legibility checks are needed to catch criteria that are formally consistent but practically wrong.

3 Pattern Specification

Let $A = \{a_1, \dots, a_k\}$ denote the active axis set. Let x_t be the current frontier state and let m be a proposed move. Each axis has a scoring or review function $s_i(\cdot)$ that is fixed for the run. UFO admits a move when

$$\forall a_i \in A : s_i(m) \geq s_i(x_t) \quad \text{and} \quad \exists a_j \in A : s_j(m) > s_j(x_t). \quad (1)$$

Dominated moves are discarded or returned for revision. They are not rescued by averaging, weighting, or confidence aggregation.

3.1 Rules

R1: strict dominance gate. Equation 1 is the sole admissibility rule. A move may not improve one axis by sacrificing another declared axis.

R2: axis-first governance. The axis set A is declared externally by a human, policy, or meta-policy before orchestration begins. UFO has no inference path from outputs to objective. Undeclared axes are outside the orchestrator’s Umwelt.

R3: no pre-specified terminal utility. There is no target scalar value and no optimality predicate. Progress is frontier movement, not goal-match.

R4: saturation-based exit, separated from budget. Orchestration exits when no pending proposal is admissible under the current axis declaration. Time, compute, and token limits are budgets; budget exhaustion is a graceful halt, not task completion.

Scalar-isolation companion clause. Confidence scores may inform prompting or review attention, but they must not enter the dominance gate. Allowing confidence to gate admissibility would reintroduce scalar aggregation through the back door and weaken the convergence-resistance hypothesis in Section 7.

3.2 Mechanism: exclusion-topology

The mechanism is a boundary process (Figure 2). Proposal agents inject pressure against a membrane. The membrane is defined by rejection: a proposal either crosses because it improves at least one axis and harms none, or it remains outside the frontier. The interior is therefore an axis-space region, not a target point.

3.3 Failure modes pinned to rules

Table 1 maps the mechanism roles to collapse paths. These are construction-level claims: removing a rule reopens a failure mode even before any external benchmark is run.

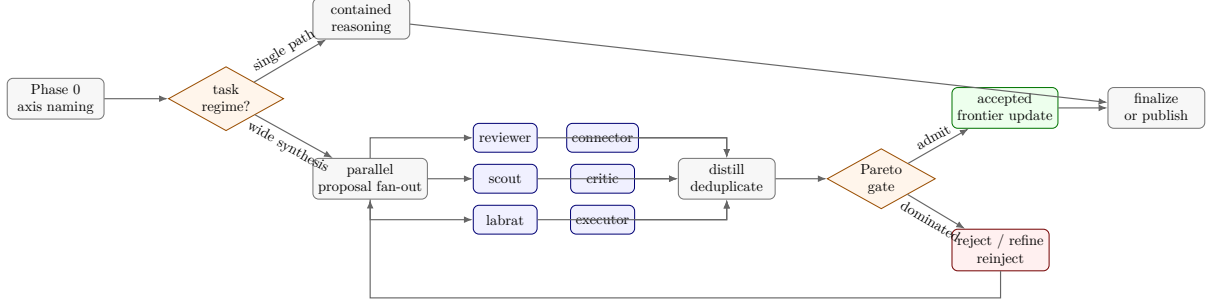


Figure 1: UFO orchestration loop. Parallel proposal generation supplies velocity; a distillation step removes duplicates and contradictions; the strict Pareto gate is the only admission membrane; saturation, not budget exhaustion, is the success exit.

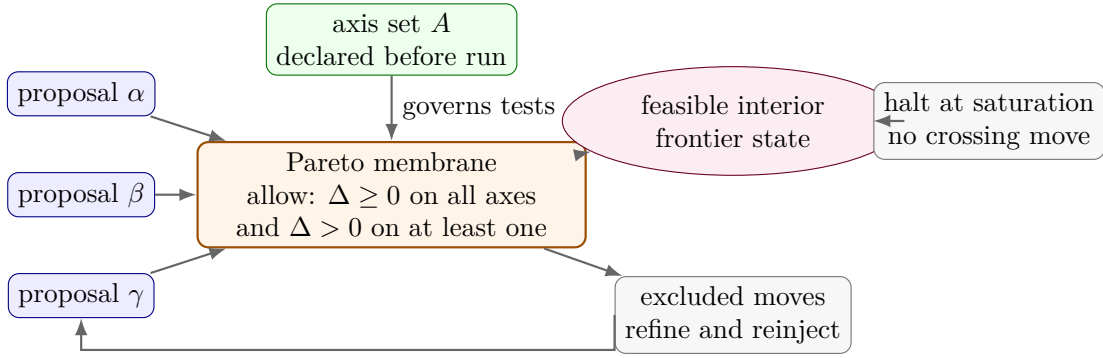


Figure 2: Exclusion-topology. The boundary is defined by what the filter rejects. Axis governance feeds into the membrane; the membrane enforces the boundary but does not infer the objective.

4 Lineage of the Assembly

The source draft frames UFO through nine lineage nodes. In the arXiv-style version, these nodes are grouped by functional role rather than presented as a loose list.

Axis drawing. Von Uexküll and de Waal supply the methodological warning that the world of the task is not identical to the evaluator’s convenience vocabulary [9, 10]. UFO operationalizes this as axis-first governance.

Boundary integrity. Antimetabole supplies a rhetorical analogy for symmetrical commitments: the same axis set and scoring functions govern both acceptance and rejection [11, 12]. Strict Pareto dominance supplies the formal gate [1, 2].

No-goal exploration. Novelty search and MAP-Elites motivate abandoning terminal scalar objectives in deceptive or under-specified landscapes while preserving structured exploration [3, 4]. The ratchet framing retained in the source draft is interpreted as monotone gain retention across stages [15].

Metacriteria. Pirsig and Scott warn that formal criteria are not self-validating [13, 14]. In UFO, this motivates audit loops for quality recognition and legibility.

Table 1: Mechanism roles and rule-linked failure modes.

Role or rule	Function	Failure mode when absent
Velocity	Heterogeneous peers inject proposal pressure.	Homogeneous convergence; frontier receives no multi-angle pressure.
Filter / R1	Strict Pareto admission blocks axis regressions.	Single-axis capture; scalarization admits trade-offs as aggregate wins.
Umwelt / R2	Axes are declared from task structure before the run.	Objective-space lock-in; axis-inference drift; wrong boundary is enforced.
No-goal / R3	No terminal scalar target is declared.	Terminal-utility collapse; search becomes scalar-reward optimization.
Exit / R4	Saturation is separated from time or token budget.	Budget-exit masquerade; fixed-horizon halt is mislabeled success.

5 Evidence Layer: Preregistered Internal Validation

This section reports the evidence layer as internal validation rather than external performance proof. The probes ask whether the mechanism’s own failure protections remain observable under ablation, pair removal, and mutation. They do not establish that UFO outperforms other orchestrators on benchmark tasks.

5.1 Single-node ablation battery

The source draft reports nine HYPOTHESIS/METHOD/RESULT triples. In each case, the hypothesis is that a lineage node is decorative; the method removes or weakens the node under controlled probe conditions; the reported result is that the node is load-bearing. Table 2 compresses the battery.

Table 2: Individual-node ablation battery. Results are internal to the preregistered probe design.

Node	Probe setting	Reported result	Failure exposed by removal
Umwelt / von Uexküll	Hidden-room grid with aliased sensor outputs.	Load-bearing	Causal aliasing and invalid generalization.
Anti-anthropocentric / de Waal	Multi-agent foraging with species-level value conflicts.	Load-bearing	Proxy brittleness and ecological constraint strain.
Antimetabole / symmetry	Paired objectives over nonlinear trade-off surfaces.	Load-bearing	Directional bias and narrowed exploration.
Pareto dominance	Many-objective deceptive benchmarks with scalar ties.	Load-bearing	Loss of middle-front stepping stones.
Novelty search	Maze-like plateaus and deceptive local optima.	Load-bearing	Plateau-optima lock.

Node	Probe setting	Reported result	Failure exposed by removal
MAP-Elites / illumination	Behavior binning and out-of-distribution transfer.	Load-bearing	Reduced behavior-mode coverage.
Ratchet	Monotone curriculum versus reset-heavy progression.	Load-bearing	Oscillation and forgetting.
Quality recognition / Pirsig	Matched numeric fitness with and without review loop.	Load-bearing	Brittle artifacts and weak repairability.
Legibility / Scott	Human-in-the-loop adaptation with opaque versus legible traces.	Load-bearing	Misinference, delayed intervention, and override risk.

5.2 Pair-removal battery

Pair removal asks whether two nodes fail independently or redundantly. The reported outcome is 9/10 complementary pairs and one redundant pair. The redundant case, antimetabole plus Pirsig, is interpreted as overlap in semantic-integrity coverage rather than evidence that either node is decorative.

5.3 Mutation kill-rate matrix

The source draft reports a Round 2 mutation pass with 13 prototype mutations and a 10/13 kill count, or 76.9%. The preregistered floor was 75%. Three surviving mutations identify single-anchor gaps: fabricated kill-rate numbers, a translation-citation anchor, and the scalar-isolation companion clause. Table 4 gives the family-level summary; Appendix A gives the compact mutation matrix.

6 Restricted Substrate Comparison: UFO+SQLite versus Honcho

The source draft includes a memory-layer comparison between a local SQLite persistence cache and Honcho. This comparison is restricted to memory-layer and token-cost axes; it is not an end-to-end orchestration comparison. The result is substrate-specific rather than universal: SQLite is favored for xbreed-like control-plane recall, while Honcho is positioned as appropriate when memory itself is a user-visible product.

The token-preservation narrative is the cleanest quantitative axis in this comparison. In the reported probes, SQLite emits raw findings while Honcho returns service-shaped summaries. The source draft reports deltas of +50% for cold-start recall, +28% for warm-round recall, +95% for cross-session recall, and +146% for a longer rubric-shaped finding. These observations support a substrate-choice principle: use local structured persistence when memory is a control-plane cache; use a memory service when memory is the product interface.

7 Preregistered Predictions

The following statements are hypotheses, not measurements. They are recorded before external benchmark evaluation to prevent post-hoc framing.

Table 3: Pair-removal summary.

Pair	Nodes	Classification	Compound failure mode
8+9	Pirsig + Scott	Complementary	Compound safety debt: quality-recognition failure plus operator-legibility failure.
1+2	Umwelt + anti-anthropocentric	Complementary	Double perceptual collapse.
4+7	Pareto + ratchet	Complementary	Oscillating frontier with no memory.
5+6	novelty + MAP-Elites	Complementary	Exploited-and-hollow search space.
3+4	antimetabole + Pareto	Complementary	Inverted solutions dominate frontier.
2+7	anti-anthropocentric + ratchet	Complementary	Irreversible anthropocentric lock-in.
1+6	Umwelt + MAP-Elites	Complementary	High-coverage archive with poor transfer.
8+4	Pirsig + Pareto	Complementary	Single-metric scalarization capture.
3+8	antimetabole + Pirsig	Redundant	No new failure beyond semantic-integrity overlap.
5+9	novelty + Scott	Complementary	Unconstrained novelty noise.

P1: premature convergence. UFO should reduce some premature-convergence failures relative to fixed-horizon debate or fixed-graph completion in incident-response-like synthesis tasks, because dominated moves fail the admissibility gate. This hypothesis depends on the scalar-isolation clause and on adequate axis design.

P2a: cloud-substrate cost. UFO should underperform in tight-cost regimes under cloud-API substrates when CLI spawn overhead is not amortized. This prediction may flip under local-model substrates.

P2b: stateful refactor tasks. UFO should underperform on stateful refactor tasks where graph-native orchestration encodes constraints directly as typed edges. This is expected to hold regardless of substrate.

P3: axis quality. Practical utility is sensitive to axis design quality. Poor axes create blind spots not detectable from inside the loop. This is a precondition for P1.

P4: shared-error cascades. Cross-model plurality will not fully prevent shared-error cascades when all peers inherit the same mistaken source assumptions. Confidence capping can reduce but not eliminate the risk.

P5: substrate half-life. The CLI-as-substrate advantage should decay at vendor-convergence events for cloud workloads, but local-model and anti-lock-in workloads may retain a long-tail advantage. This prediction is entangled with P2a.

Table 4: Mutation family summary. Kill rate is a specification-internal consistency measure, not an external performance metric.

Mutation family	Killed / total	Interpretation
Four-rule flips	3/3	Core rules had explicit collapse anchors.
Discriminator row swaps	2/2	Halting-sensitivity ranking was anchored in two locations.
Prediction hedge removal	2/2	Preregistration language was cross-anchored.
Limitations silent drop	2/2	Limitations were linked to prediction dependencies.
Kill-rate fabrication	0/1	Numeric prior baseline lacked a second anchor.
Two-layer violation	1/1	Pattern/evidence firewall caught misplaced empirical language.
Citation integrity	0/1	Translation citation needed an independent anchor.
Rule enforcement	0/1	Scalar-isolation clause needed a prediction back-reference.

8 Limitations

Axis-naming bias. R2 is load-bearing because a strict gate over bad axes is still a bad gate. UFO cannot see undeclared dimensions from inside the loop.

Cross-model collusion. Heterogeneous vendors do not guarantee independent beliefs. Shared source assumptions can cause aligned errors across peers.

Substrate half-life. CLI-mediated orchestration may lose distinctiveness as vendor protocols converge. The predicted decay is event-triggered rather than smooth.

Unknown-objective tasks. UFO requires named axes. It is poorly matched to tasks in which the objective is genuinely unknown and cannot be responsibly externalized before the run.

Sequential-depth tasks. Wide fan-out is not a general substitute for deep sequential reasoning. Task-regime classification is therefore part of the pattern rather than a mere engineering optimization.

9 Future Work

The next step is external evaluation against comparable systems. The source draft frames this as a milestone sequence. First, run a single-task baseline to test axis-quality assessment and premature-convergence behavior. Second, evaluate paired task regimes: wide multi-axis synthesis versus contained sequential reasoning. Third, compare against graph-native and debate-native orchestrators under matched budgets. Fourth, add quality-diversity baselines such as MOME before making claims about assembly from existing primitives [5]. Finally, convert internal reports into public supplementary artifacts with stable identifiers.

Table 5: Restricted substrate comparison. “Unverified” means the source draft reports that public documentation did not expose the corresponding numeric observable.

Axis	UFO+SQLite observable	Honcho observable in source draft
Cold start	Local file open and SQL read; measured in milliseconds on local disk.	Network call and token validation; numeric latency not published.
Write latency	WAL transaction for a single finding; local, idempotent schema.	Immediate message add with asynchronous processing; numeric latency not published.
Read latency	B-tree lookup for recent findings; no network or auth.	HTTP retrieval plus JSON parsing and semantic recall; read RTT not published.
Tokens per recall	Raw row text injected into context; baseline around 125 tokens in simulated probes.	Summary or dialectic-reasoning wrapper; reported overhead ranges from +28% to +146% depending on scenario.
Cross-session coherence	File read by any process with filesystem access; no session reinitialization.	Session get-or-create and summary cadence impose service-mediated continuity.
Lock-in surface	Portable SQLite schema and dump/migration path.	Proprietary API surface and externally versioned contract.
Degradation	Missing database can degrade to empty recall under the Phase 0 contract.	Network, auth, or service failure can block memory calls; degradation contract not documented in the source draft.

10 Conclusion

UFO is best read as a disciplined coordination topology rather than as a universal agent architecture. Its novelty claim is compositional: heterogeneous proposal pressure, externally declared axes, strict Pareto admissibility, and no terminal scalar utility are combined into a boundary process. The paper’s current evidence is deliberately modest. It shows how the mechanism is meant to fail when rules are removed, and it records preregistered hypotheses for future benchmarks. That restraint is part of the contribution: it keeps the pattern claim, the evidence claim, and the future performance claim separable.

Acknowledgments

This arXiv-style manuscript was derived from the uploaded source draft and preserves its key caveats: several citations are marked as unverified or partially verified in the source, and the reported probes are internal validation artifacts rather than external benchmark results.

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A Compact Mutation Matrix

Table 6: Round 2 mutation matrix condensed from the source draft.

ID	Target	Prediction	Actual	Catch mechanism or surviving gap
M1	R1 strict improvement weakened to no-worse only	Kill	Killed	Single-axis capture anchor.
M2	R2 axis declaration moved into run	Kill	Killed	Objective-space drift anchor.
M3	R4 saturation replaced by budget exhaustion	Kill	Killed	Budget-exit masquerade anchor.
M4	Halting-sensitivity ranking fully inverted	Kill	Killed	Two-location ranking anchor.
M5	M1/M3 ranking label swap	Kill	Killed	M1-sharpest assertion.
M6	Preregistration marker removed	Kill	Killed	Multiple preregistration anchors remain.
M7	Dependency chain removed	Kill	Killed	Section 5 and Section 7 cross-anchor.
M8	Axis-naming limitation removed	Kill	Killed	P3 dependency becomes orphaned.
M9	Cross-model collusion limitation removed	Kill	Killed	P4 limitation pairing exposes removal.
M10	Kill-rate baseline fabricated	Kill	Survived	Numeric prior baseline had only one anchor.
M11	Empirical result inserted into pattern rule	Kill	Killed	Pattern/evidence firewall.
M12	von Uexküll translator changed	Kill	Survived	Citation needed an independent anchor.
M13	Scalar-isolation companion clause deleted	Kill	Survived	P1 needed an explicit back-reference.

B Artifact Register

The source draft references the following non-public or project-local artifacts as preregistration or audit records: a project plan, a Pareto/prior-art scan, an orchestration prior-art scan, a nine-node lineage battery, a pair-removal battery, a mutation battery, and substrate stress reports. A public arXiv submission should either include these as supplementary material or replace path-local citations with stable archival identifiers.